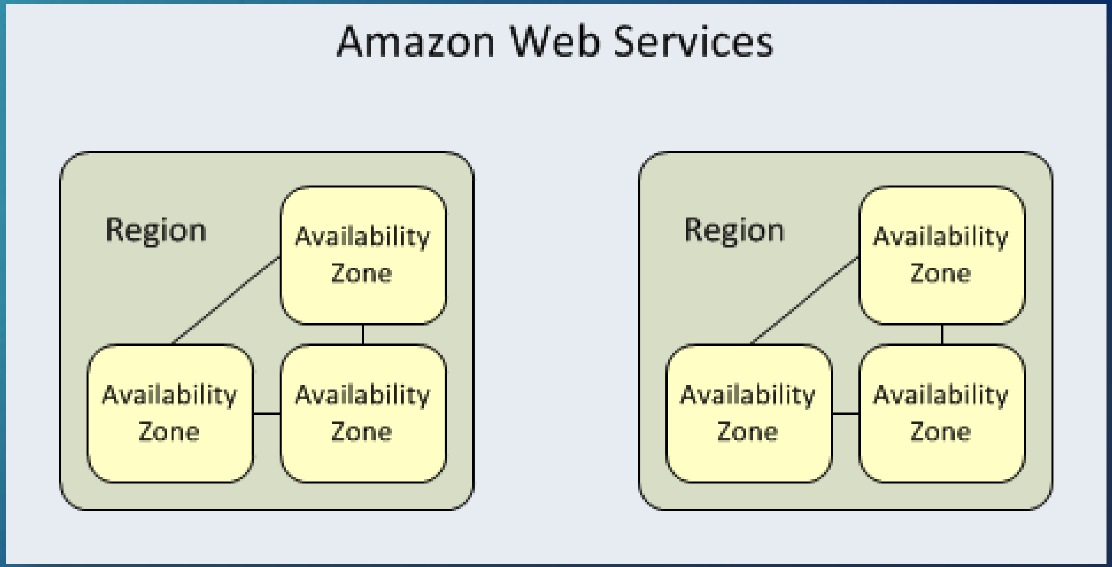
RECAP

- S3 buckets are tied to a specific AWS Region
  - True ✅
  - False
- Which AWS storage option is the most cost-effective for storing large amounts of static data that doesn’t require frequent access?
  - EBS
  - EFS
  - S3 ✅
  - Instance Store
- An EBS volume persists independently from the running life of an EC2 instance
  - True ✅
  - False

RECAP

- A <u>Region</u> contains 2 more or <u>Availability Zones</u>
- Regions
  - Separate geographic area where your EC2 instances are running
  - Each is designed to be completely isolated from the other to provide fault tolerance and stability
- Availability Zones
  - Isolated, but other zones in same region are connected



RECAP

- Storage models
  - Simple Storage Service (S3)
    - Cheapest for data storage and can be accessed from anywhere via a URL
  - Elastic Block Store (EBS)
    - Mount to a single EC2 instance
    - Only available in a particular region, but faster than S3 and EFS
  - Elastic File System (EFS)
    - Available on <u>multiple regions</u> and can be accessed by up to *thousands* of EC2 instances at once
    - Best for large very large quantities of data such as large analytic workloads
  - Other specialized models:
    - Glacier, Storage Gateway, Snowball, Snowball Edge and Snowmobile

RECAP

- For your S3 website, how much do you think it costs to host per month?
  1. $1-$3/month
  2. $5-$10/month
  3. $10-$20/month
  4. > $20/month
- Answer
  - Typically, it will cost $1-3/month if you are outside the AWS Free Tier limits
  - If you are eligible for AWS Free Tier and within these limits, hosting your static website will cost around $0.50/month

AWS Advanced

Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology

Homework Assignment and Team Project

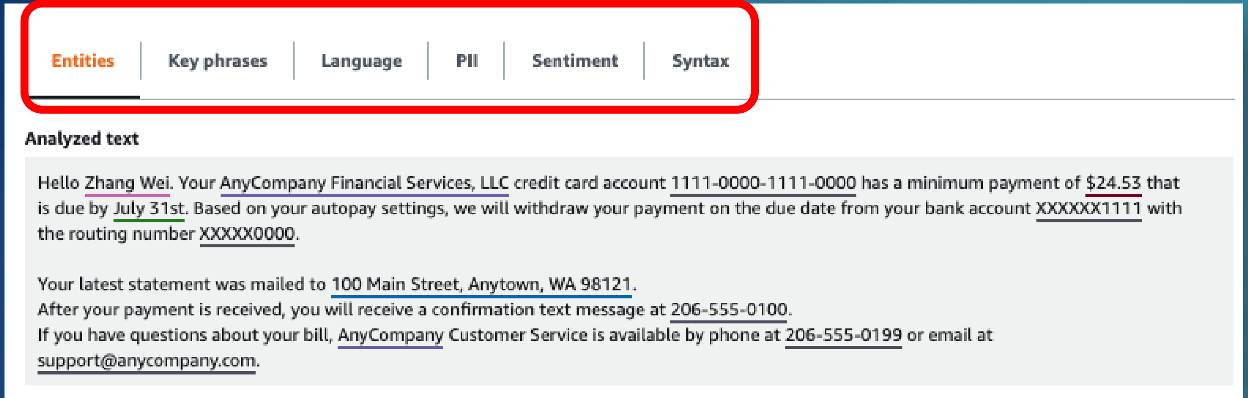
- You will research AWS tutorials of your choice and these will later be shared with the class
- The purpose of the tutorials is to:
  - Get you and your team familiar with AWS services faster to help spark ideas for the team project
  - Share your research with the class to help them better understand technologies they may not have looked at
- We will explore several AWS services in future classes but let's look at a few that won't be covered in class that you may want to research further for your project
  - Several teams used these services on past projects

AWS Comprehend

Amazon Comprehend

- A natural-language processing (NLP) service that uses machine learning to uncover information in unstructured data

Input



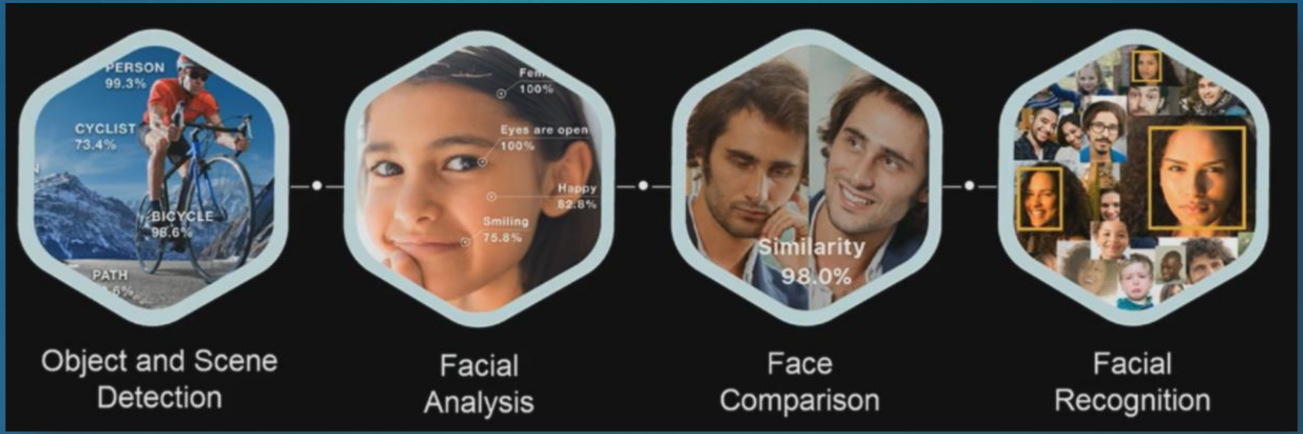
Output

Entity	Type	Confidence
Zhang Wei	Person	0.99+
AnyCompany Financial Services, LLC	Organization	0.99+
1111-0000-1111-0000	Other	0.87
$24.53	Quantity	0.99+
July 31st	Date	0.99+
XXXXXX1111	Other	0.85
XXXXX0000	Other	0.83
100 Main Street, Anytown, WA 98121	Location	0.99+
206-555-0100	Other	0.99+
AnyCompany	Organization	0.97

- Project Ideas:
  - Process a data feed to find out what people are saying about a certain entity (person, topic, etc.)
  - Analyze other content feeds (e.g. News) to look for relationships between common entities

AWS Rekognition

- Identify objects, people, text, scenes, and activities in images and videos, as well as detect any inappropriate content



- Project Ideas
  - Process a live or recorded image feed to determine certain characteristics:
    - How many people are in the image?
    - What other things are in the image (e.g. animal)?

AWS Transcribe

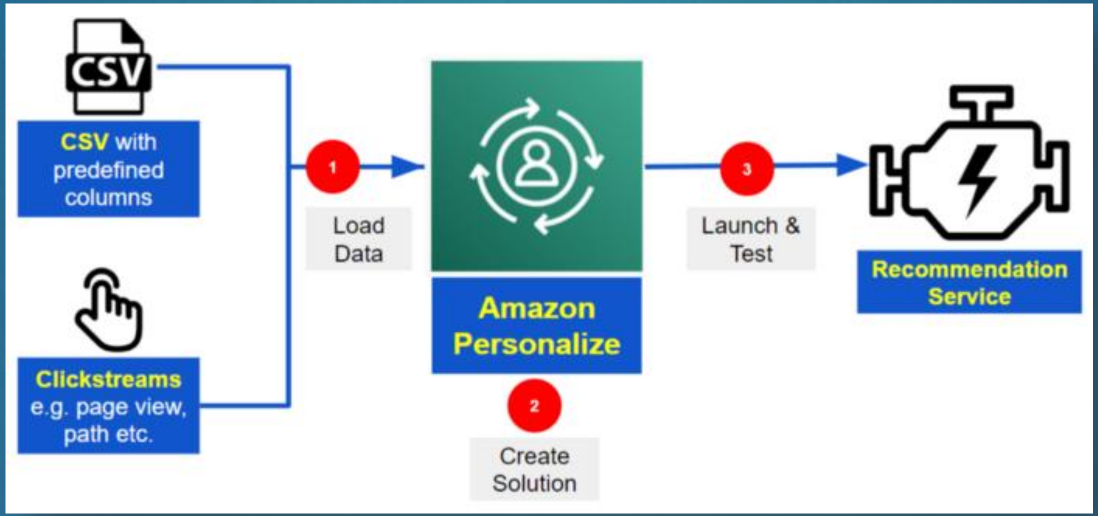
- Uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately
- It is designed to process audio input from a variety of sources such as microphones, audio or video files and provide high quality transcriptions for search and analysis

- Project Ideas:
  - Process recorded audio feed (e.g. lecture) and use the output to create a summarization
  - Process a live audio feed and search for key entities using AWS Comprehend

AWS Personalize

Amazon Personalize

- Enables developers to build applications with the same machine learning (ML) technology used by Amazon.com for real-time personalized recommendations – no ML expertise required

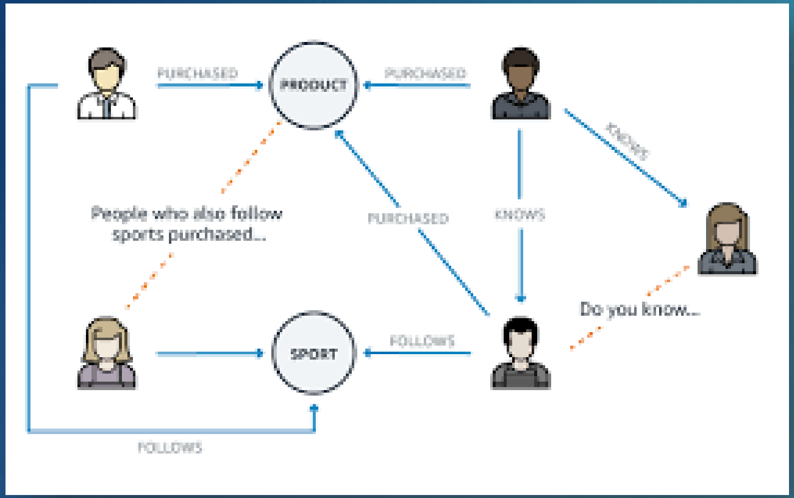


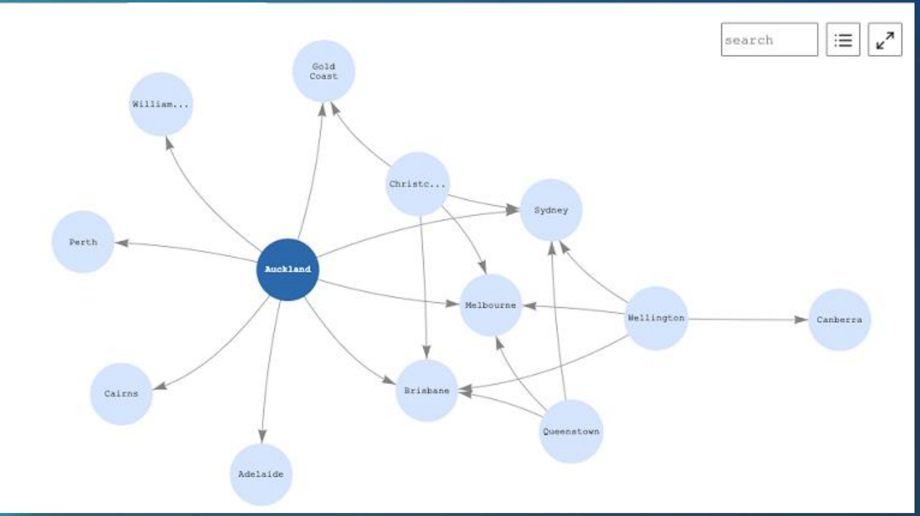
- Project Ideas:
  - Create your own specialized recommendations web site (eStore, travel, restaurant, etc.)
  - Process a resume and recommend relevant jobs using a publicly available Jobs API

AWS Neptune

Amazon Neptune

- Fully managed graph database service that makes it easy to build and run applications that work with highly connected datasets





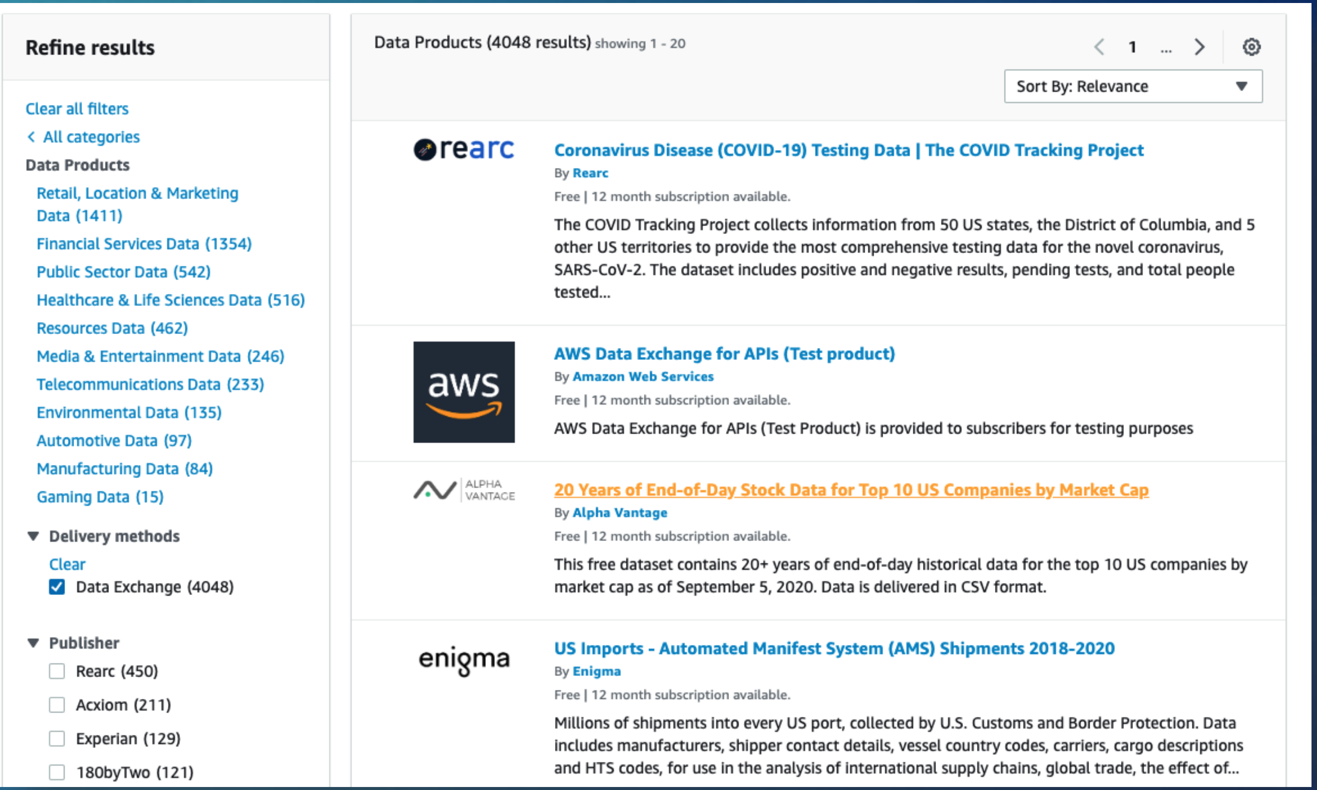
- Project Ideas:
  - Process a data feed and extract common entities (via AWS Comprehend) to visually display a graph
  - Using a data feed of locations of airports, determine the possible routes from one city to another

Other Technologies to Explore

- **AWS Amplify**
  - A set of tools and services that can be used together or on their own, to help front-end web and mobile developers build scalable full stack applications
  - This is not permitted for any class projects, but feel free to learn about this
- **Amazon Polly**
  - Service that turns text into lifelike speech, allowing you to create applications that talk, and build entirely new categories of speech-enabled products
- **Amazon Translate**
  - A neural machine translation service that delivers fast, high-quality, affordable, and customizable language translation
- **Amazon Lex**
  - A service for building conversational interfaces into any application using voice and text

Need Data?

- Check out: https://aws.amazon.com/data-exchange/



- Outside of AWS, there are many more “free open data” sets available

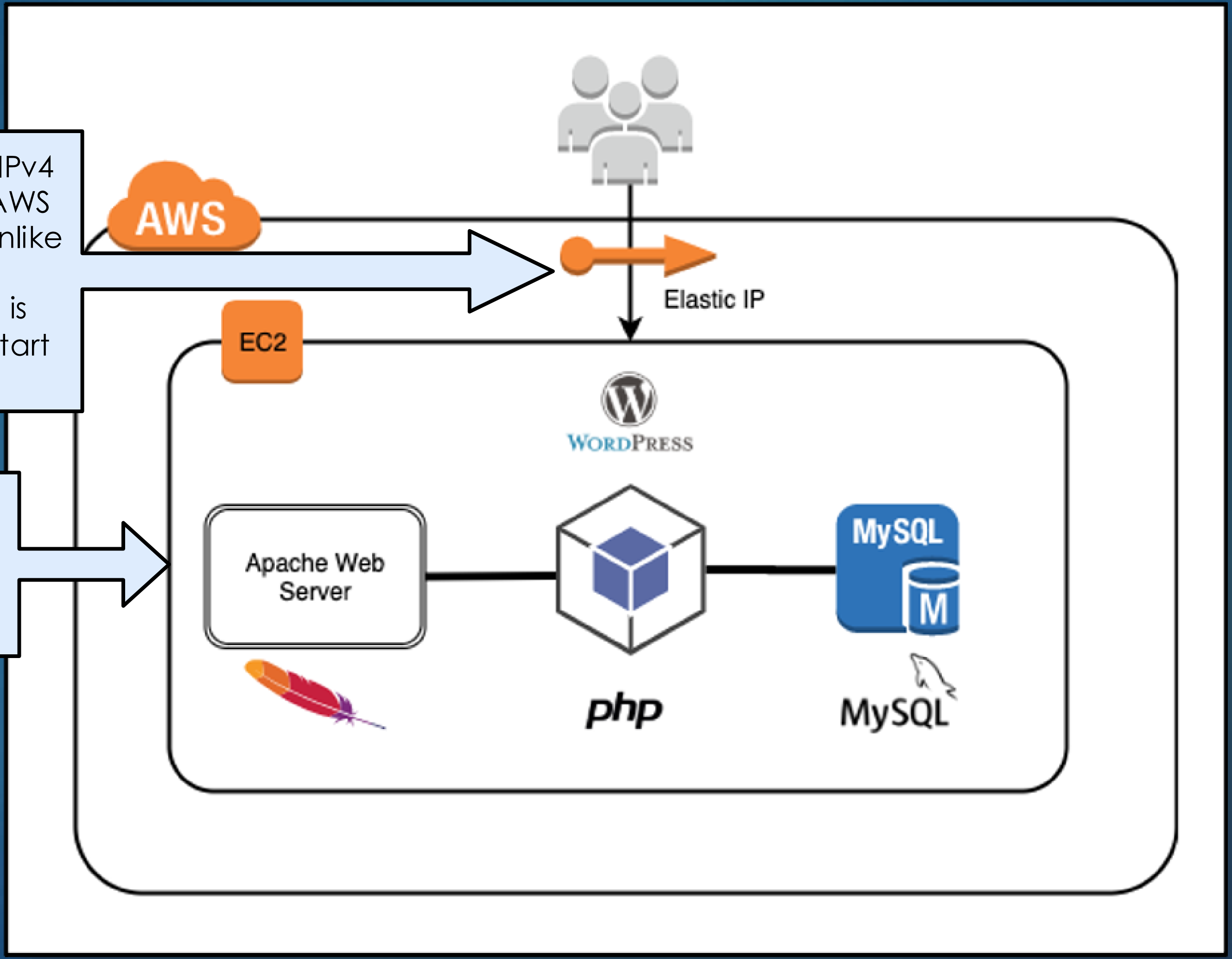
Term Project

- Either choose your own or select from the pre-defined Project Templates
- Choose your own:
  - Teams will come up with their own idea on a project using AWS services. This option offers the most flexibility for teams that want to learn and explore AWS services of their choice
- Select a Project Template:
  - Teams will select 1 of the provided project templates, which includes a high-level overview description of a project including the AWS technologies that should minimally be used
- Note that all projects must be pre-approved by Instructor (check Syllabus for date)
- In addition, teams must incorporate at <u>least one additional</u> AWS service of their choice
- Project Details: myCourses (see Helpful Links – "Term Project")

Last Activity – Install WordPress on EC2

An Elastic IP address is a static IPv4 address associated with your AWS account in a specific Region. Unlike the auto-assigned public IP address, an Elastic IP address is preserved after you stop and start your instance

Whenever you start/stop an EC2 instance, AWS will auto-assign a public IP address

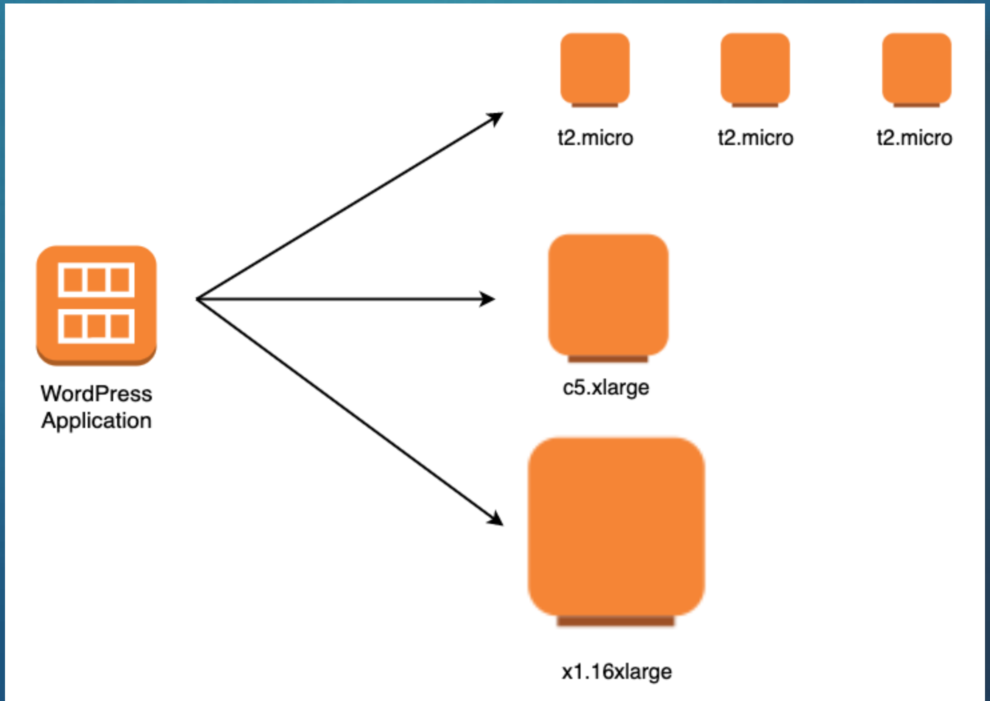


Questions:

- What if I wanted to upgrade to a larger EC2 instance?
- How can I share my EC2 instance with others?
- How could something like this be developed across a team?
- How does AWS secure my EC2 instance?
- What if I want some of my AWS services (e.g. database) to not be accessible from the Internet?

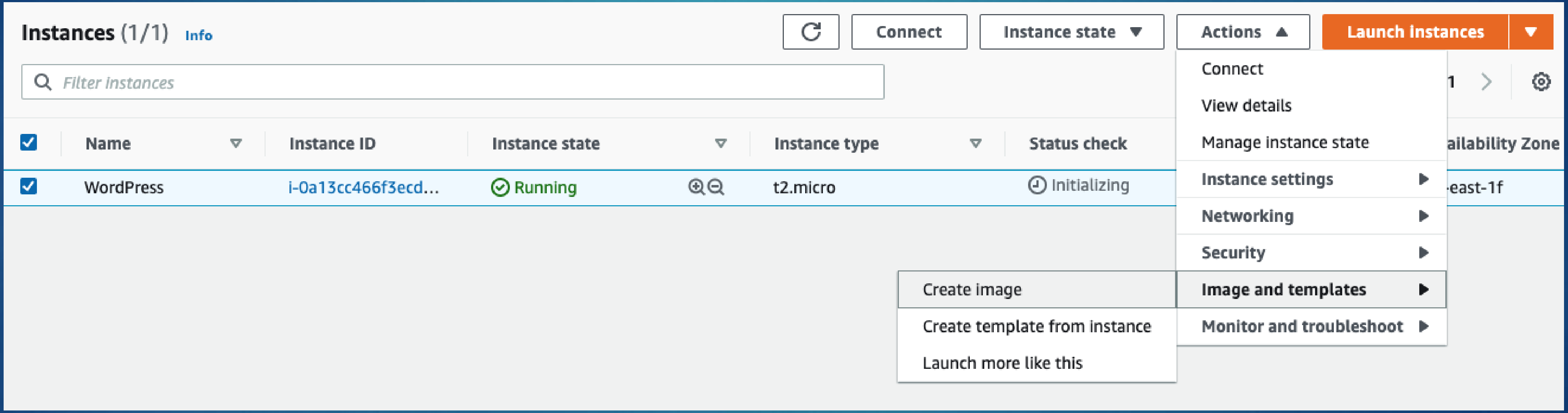
Amazon Machine Image (AMI)

- AMI is the "Golden Master"
- It provides the information required to launch an instance of any size
- You can launch multiple instances from a single AMI when you need multiple instances with the same configuration



Creating an AMI Image

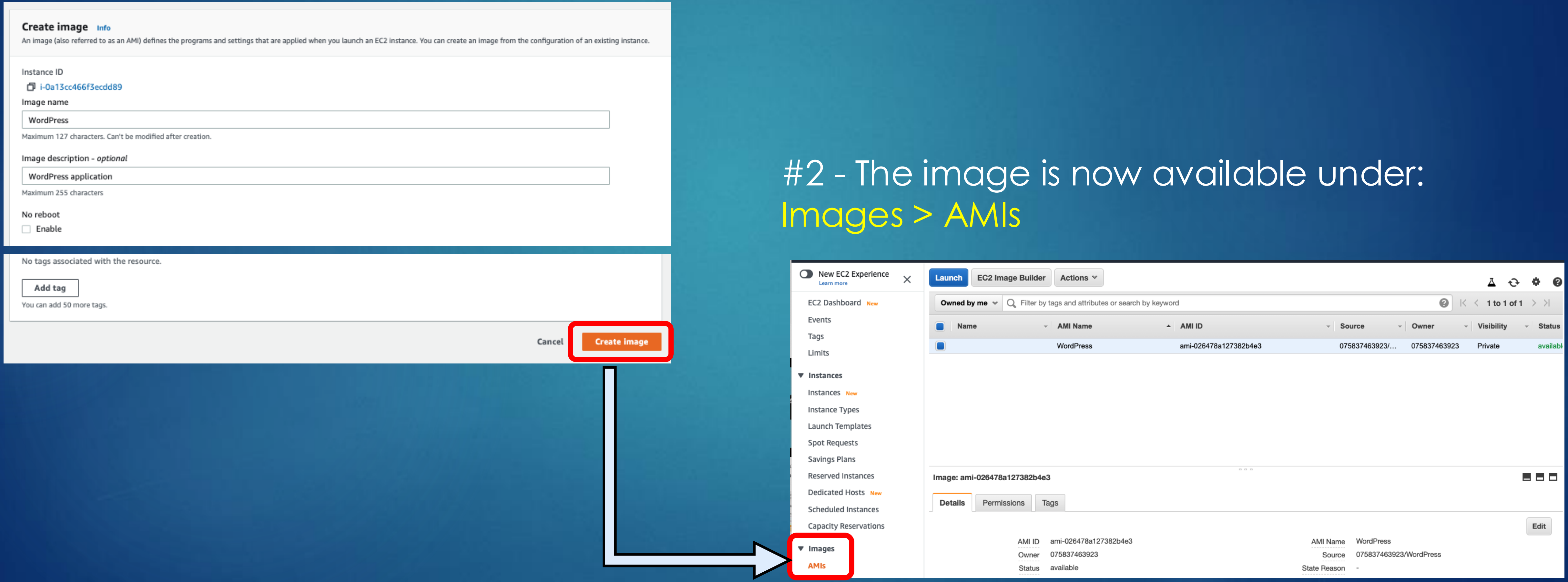
- Using your completed WordPress EC2 instance, you can create an AMI image that can be reused
- To do this, select Actions > Images and Templates > Create Image



Creating an AMI Image

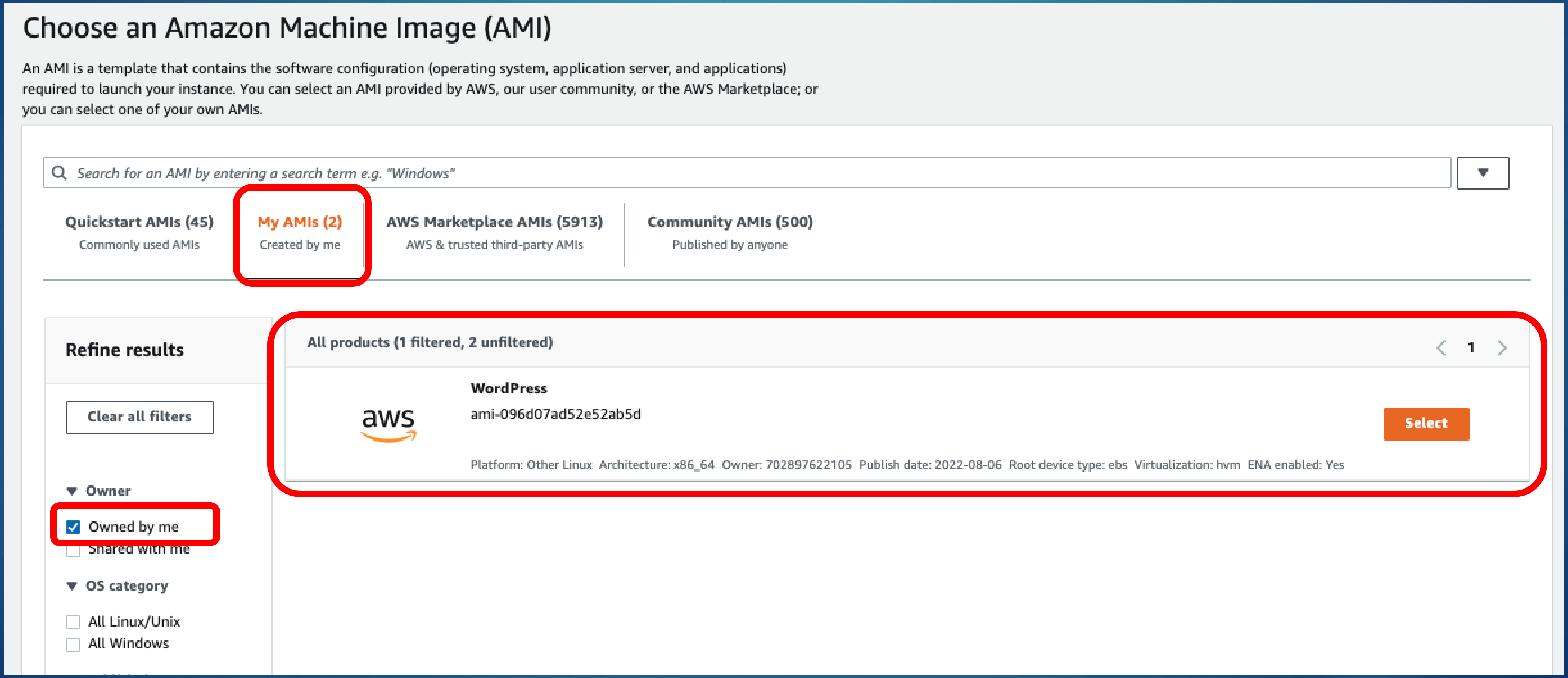
#1 - Name your image and select "Create image"

#2 - The image is now available under: Images > AMIs



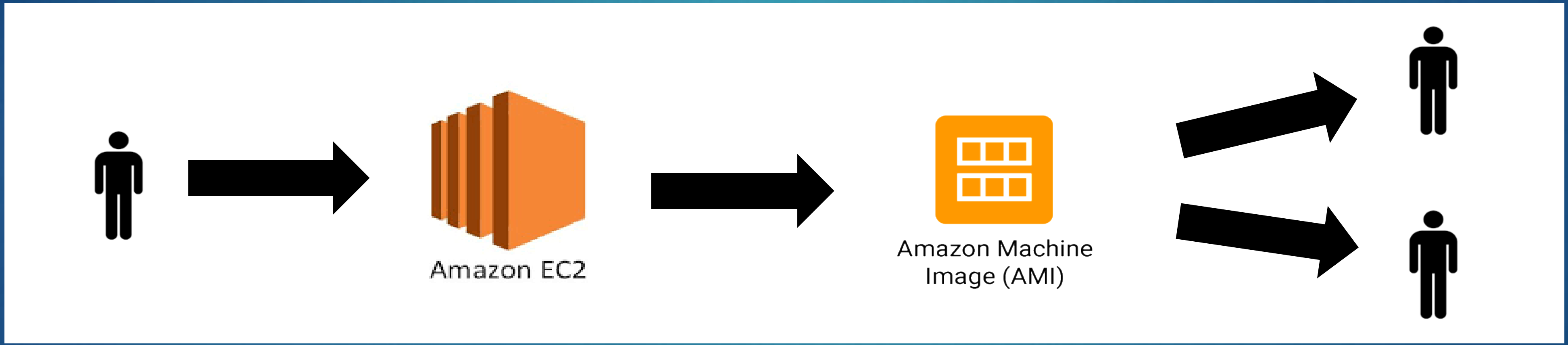
Creating an AMI Image

- Your image is now available under “My AMIs” when you configure an EC2 instance to launch



EC2 Instances - Sharing

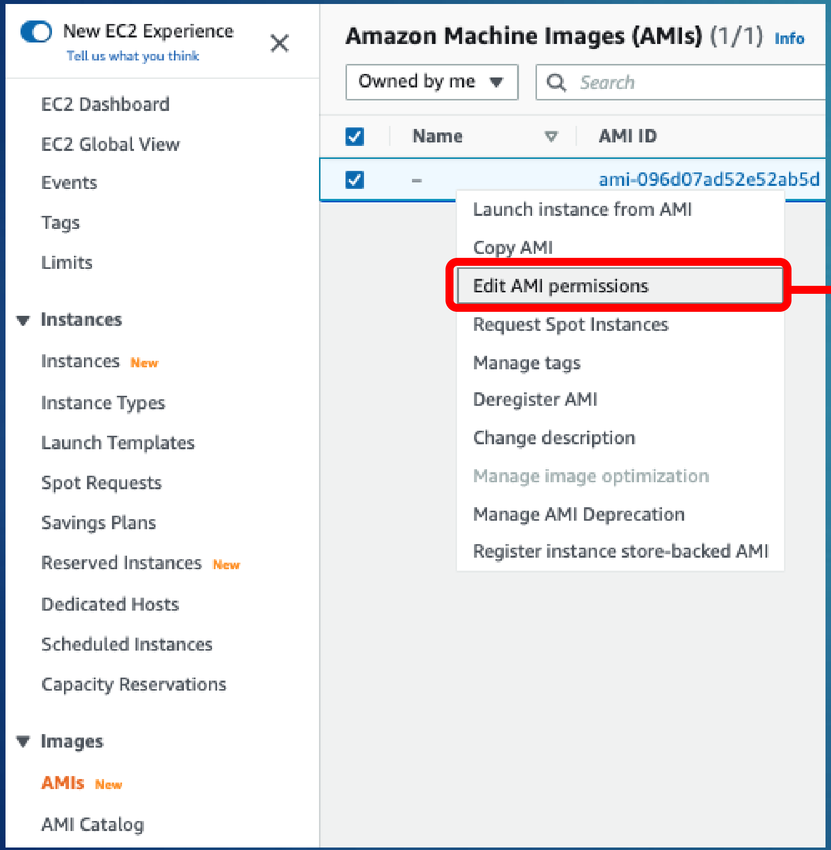
- So, if you are building an application on an EC2 instance, can you share this with your team members?
  - You cannot share a specific EC2 instance with your team



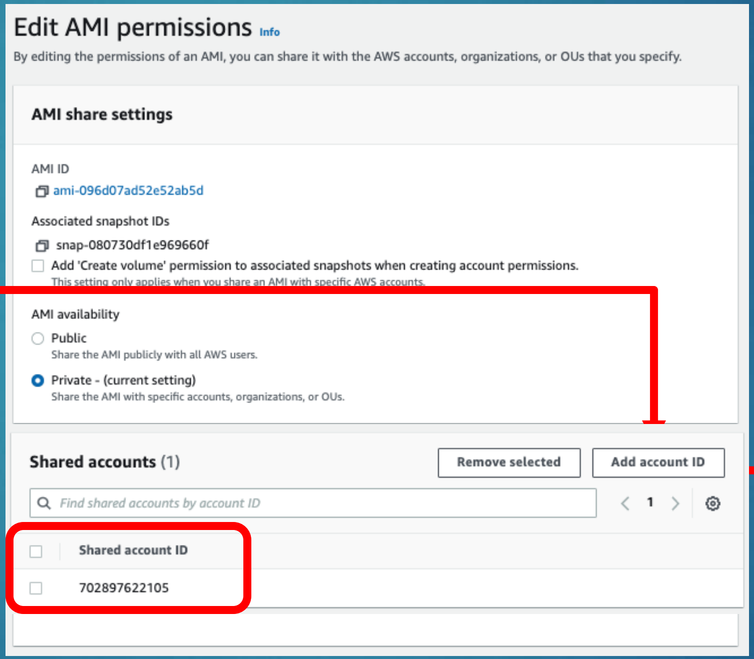
- You can share an AMI image

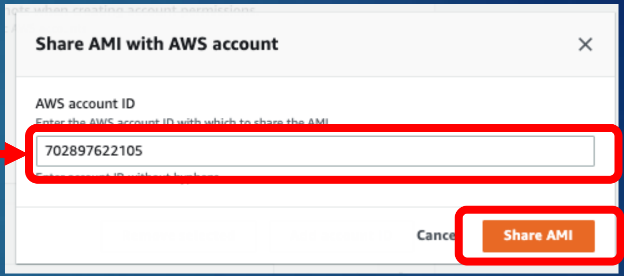
Sharing an AMI Image

- Under AMI, Select AMI image and select Edit AMI permissions



- Click "Add account ID" and add the account # of your team members and click "Share AMI"

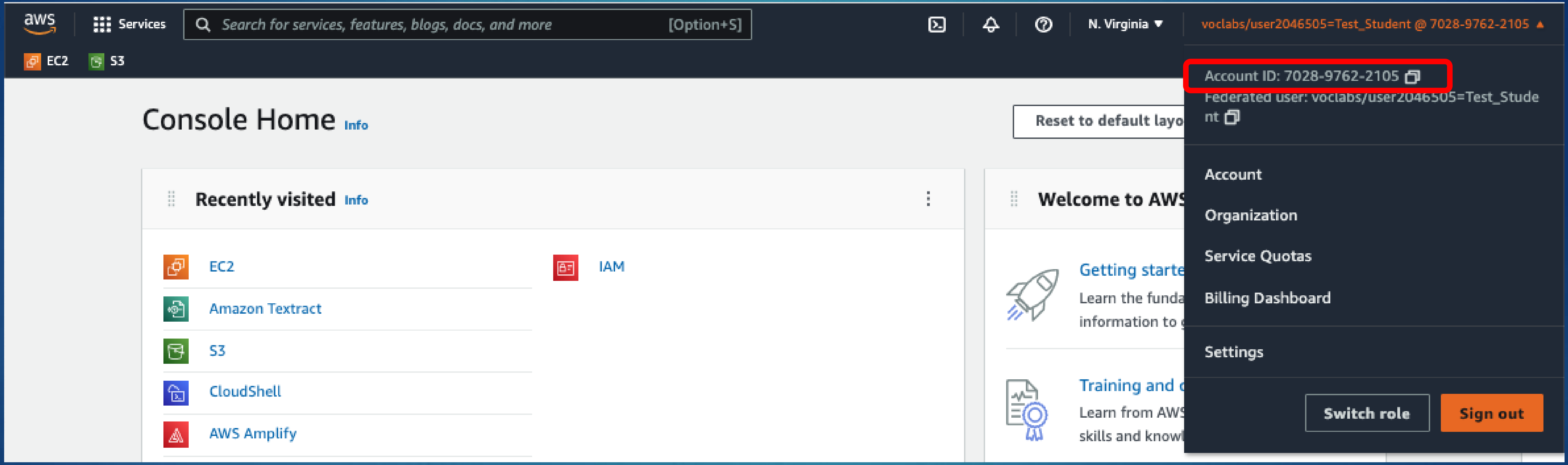




- The account ID will now appear under Shared accounts

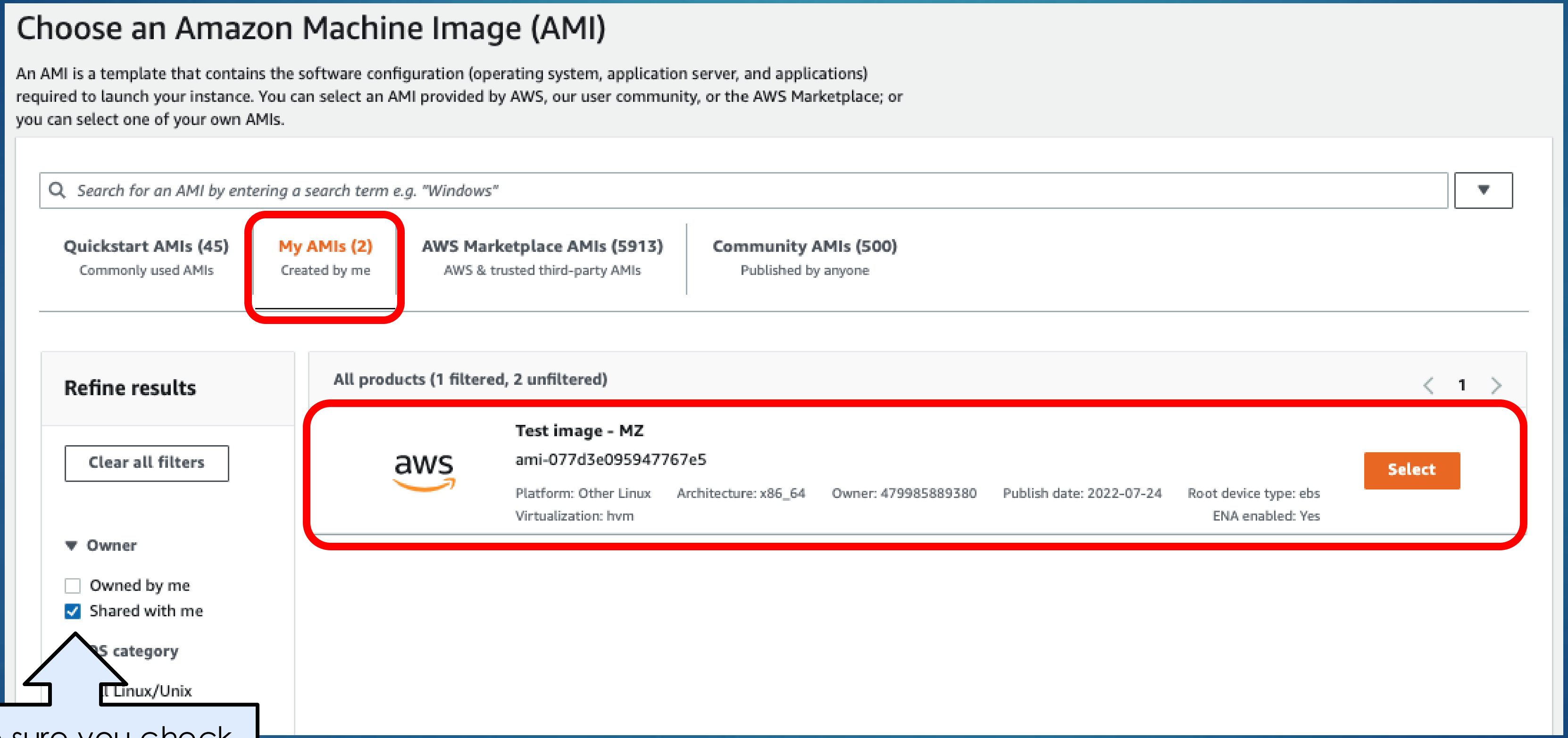
AWS Account ID

- Your account ID is in the upper right corner



Create Instance with a Shared AMI

- Team members that had their account # added, the shared image will appear under “My AMIs” when you create an EC2 instance



Make sure you check the “Shared with me” checkbox

AMI and OS

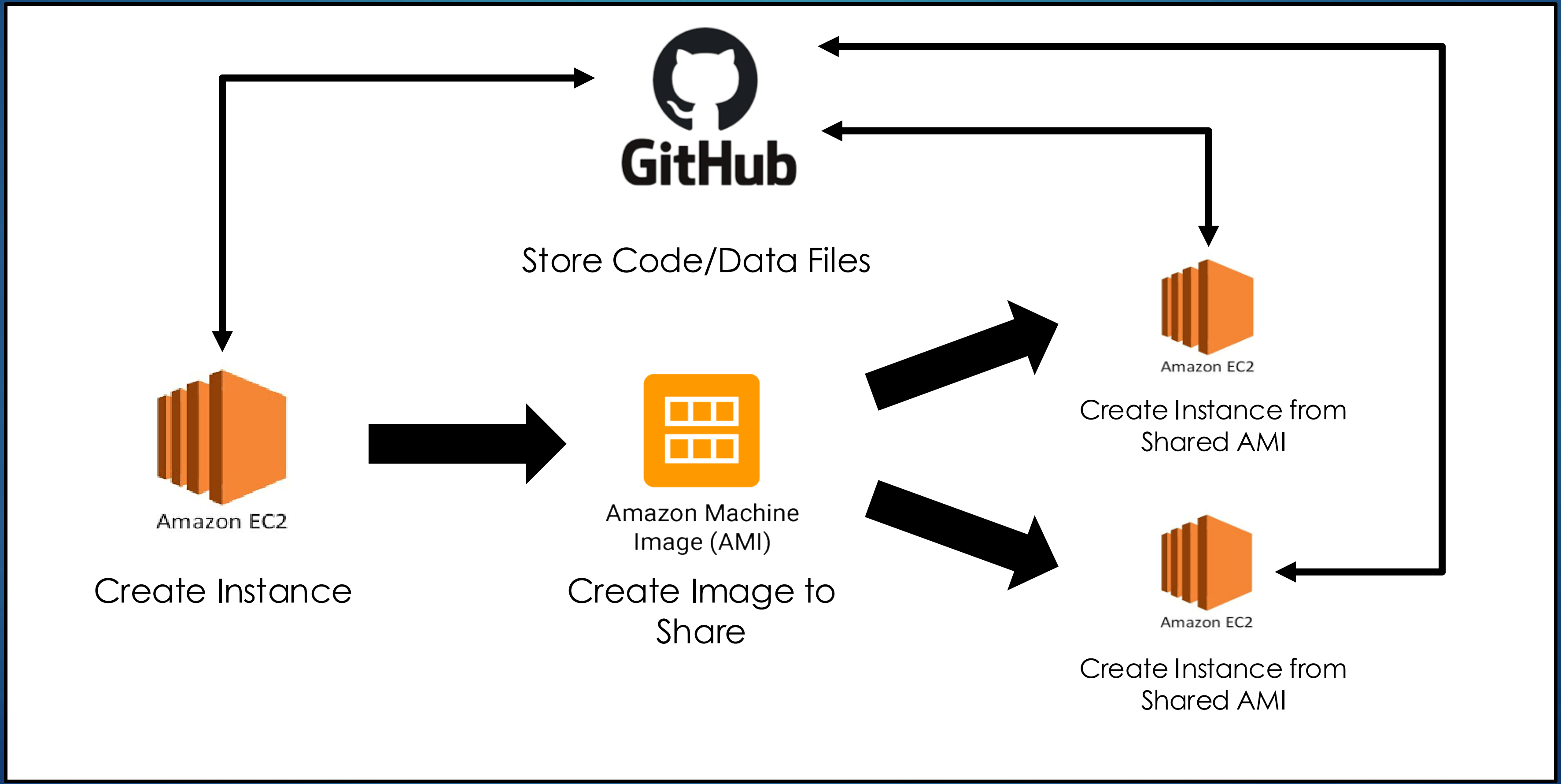
- Available in Windows, Linux and Mac
  - In class, will be working exclusively in Linux
  - For your team project, you can pick whatever OS you want, but be aware they are all not free
- There are several Linux OS available
  - These include Redhat, Ubuntu and SUSE
- We have been using Amazon Linux AMI
  - Integrates easier with AWS services and tools
  - Optimized for use in Amazon EC2 with a latest and tuned Linux kernel version

EC2 Instances – Watch your usage!

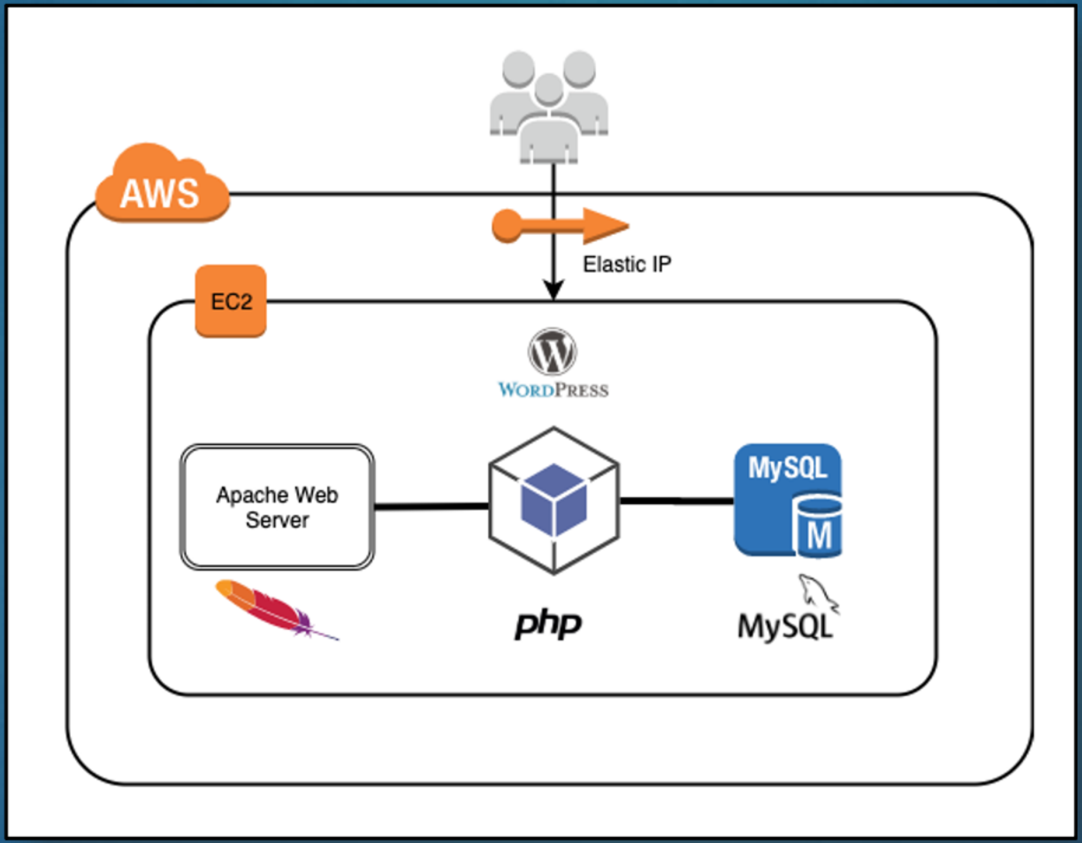
- You will get charged even if you are using it
- Only terminate if you do not plan to use your instance any more
- If you want to save your work on your EC2 instance, you can stop it and restart later
  - You will not be charged for the instance while in a stopped state
  - You will be charged other resources that are connected to it (e.g. EBS)

Sharing Code and Instances

- Share your EC2 instance via AMI for other team members to use
- Use GitHub to store all team's code and data

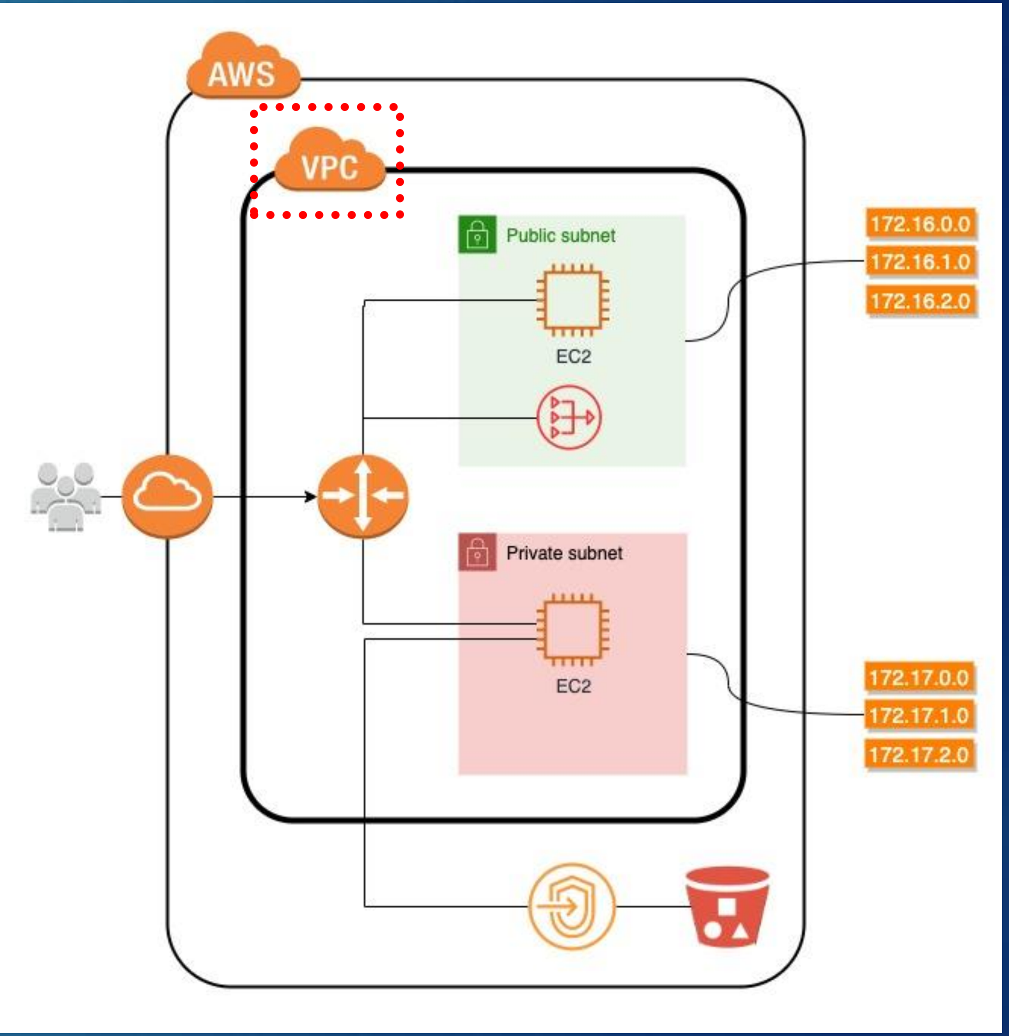


How does AWS secure my EC2 instance?



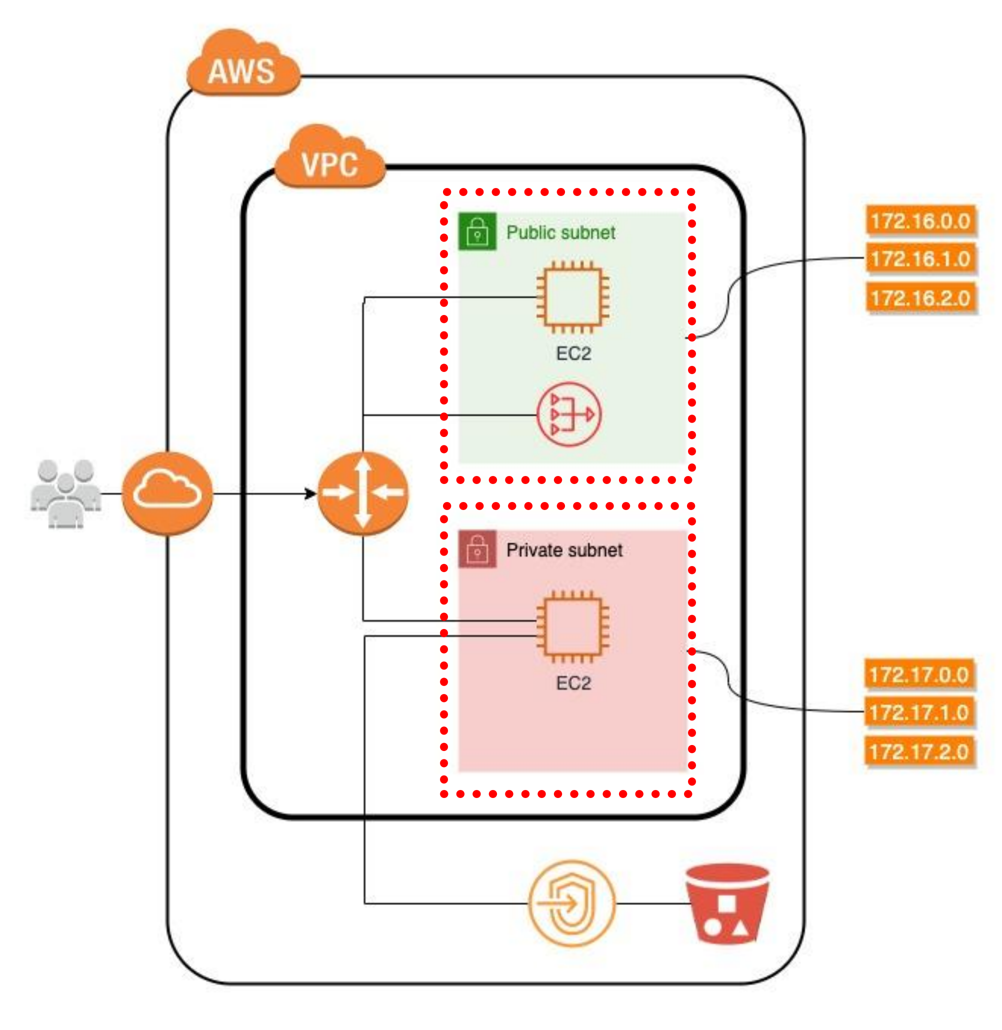
Virtual Private Cloud (VPC)

- A VPC is a virtual network dedicated to your AWS account
- Logically isolated from other virtual networks in AWS
- You can launch your AWS resources, such as Amazon EC2 instances, into your VPC
- You can specify an IP address range for the VPC, add subnets, associate security groups, and configure route tables



Subnet

- A Subnet is a range of IP addresses in your VPC
- Use a public subnet for resources that must be connected to the internet
- Resources in a private subnet can only access the internet using a network address translation (NAT) gateway that resides in the public subnet
- You can assign a subnet to a specific IP address, but it is not required
  - If you don't, AWS will automatically address it from the IP address range you assigned to it
- There is a limit of 200 subnets that can be created per VPC



Calculating IP addresses for VPC vs. Subnets

- When you create a VPC, you must specify an IPv4 CIDR (Classless Inter-Domain Routing) block for the VPC. Examples:
  - 10.0.0.0/24 ➔ $2^{32} – 2^{24} = 2^{8}$ = **256** IP addresses for your VPC (10.0.0.0 – 10.0.0.255)
  - 10.0.0.0/20 ➔ $2^{32} – 2^{20} = 2^{12}$ = **4096** IP addresses for your VPC (10.0.0.0 – 10.0.15.255)
  - The allowed block size is between a /16 netmask (65,536 IP addresses) and /28 netmask (16 IP addresses)
- After you define your VPC, you define your subnets, which are within the range of the VPC. For example, a VPC with 2 subnets:

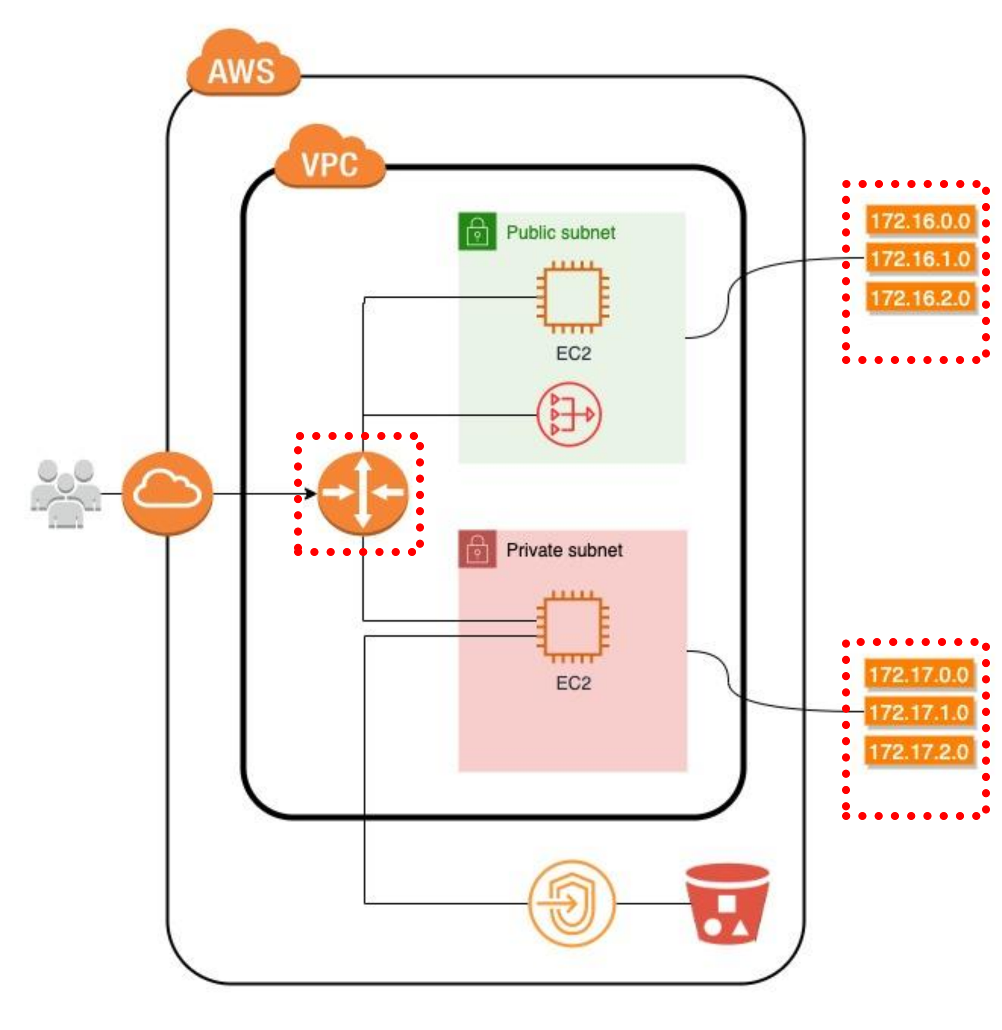
Network Component	CIDR	# of IP addresses	IP Range
VPC	10.0.0.0/24	256	**10.0.0.0 – 10.0.0.255**
Subnet #1	10.0.0.0/26	64	10.0.0.0 - 10.0.0.63
Subnet #2	10.0.0.64/26	64	10.0.0.64 - 10.0.0.127

Route Table

- A route table is a database that contains rules, or routes, that determine how traffic is directed within a Virtual Private Cloud (VPC)
- The rules specify which network traffic is directed to which network interface
- Each route in a table specifies a Destination and a Target
- For example, to enable your subnet to access the internet through an internet gateway, add the following route to your subnet route table:

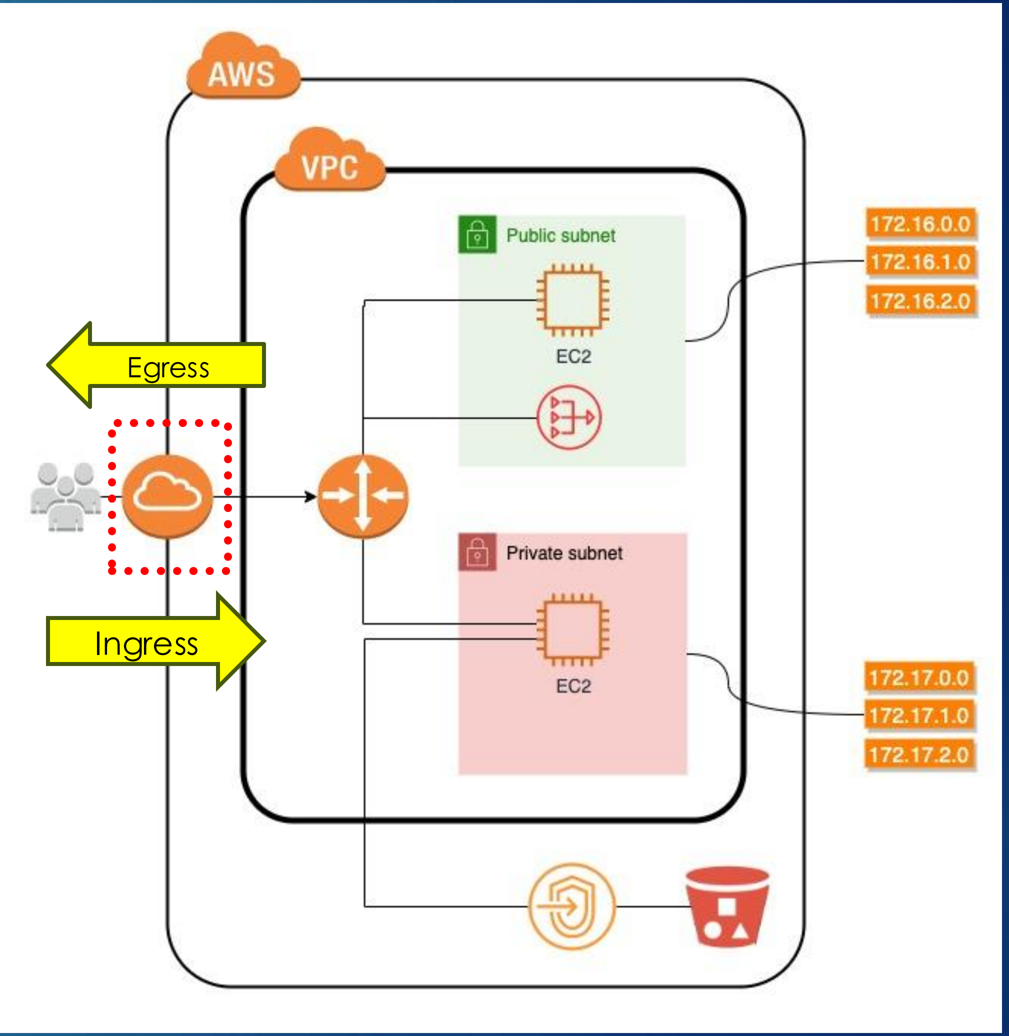
Destination	Target
0.0.0.0/0	igw-12345678901234567

- The **destination** for the route is 0.0.0.0/0, which represents all IPv4 addresses
- The **target** is the internet gateway that's attached to your VPC



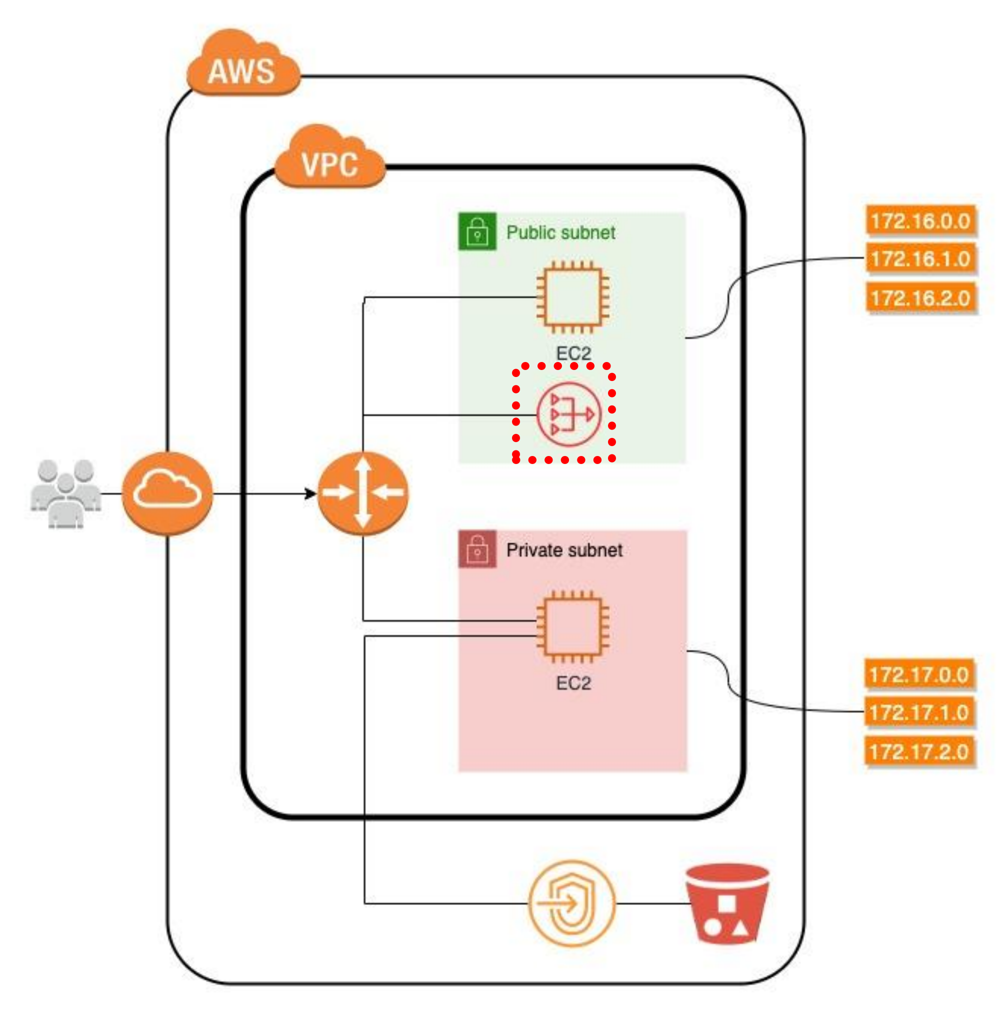
Internet Gateway

- An Internet Gateway (IGW) allows communication between instances in a VPC and the internet
- It acts as a "bridge" between a VPC and the internet, allowing you to route internet traffic to your instances
- Each VPC has only one IGW
- Ingress
  - Refers to data entering a system or network
- Egress
  - Refers to data leaving a system or network



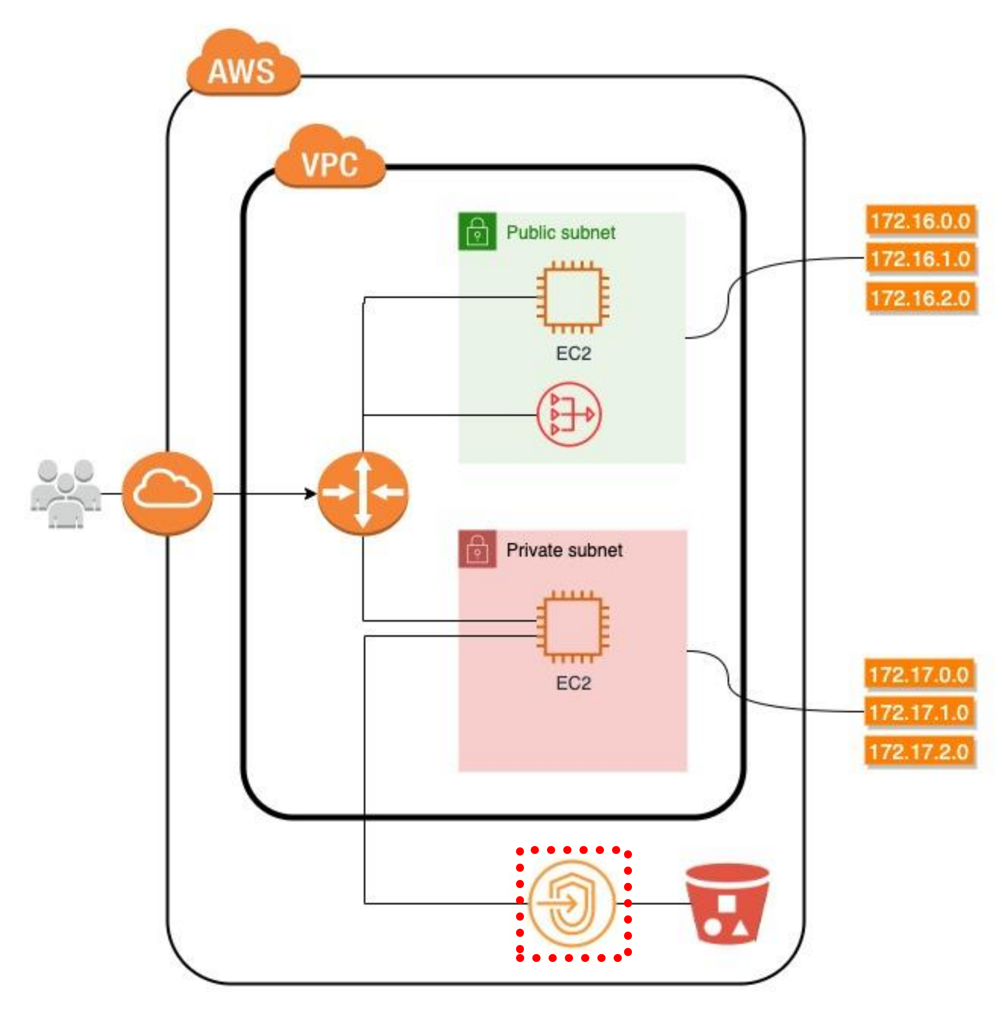
Network Address Translation (NAT) Gateway

- A NAT Gateway is used to enable instances present in a private subnet to help connect to the internet or AWS services
- It also makes sure that the internet doesn't initiate a connection with these instances
- To create a NAT gateway, you must specify the public subnet in which the NAT gateway should reside
- You must also specify an Elastic IP address to associate with the NAT gateway when you create it

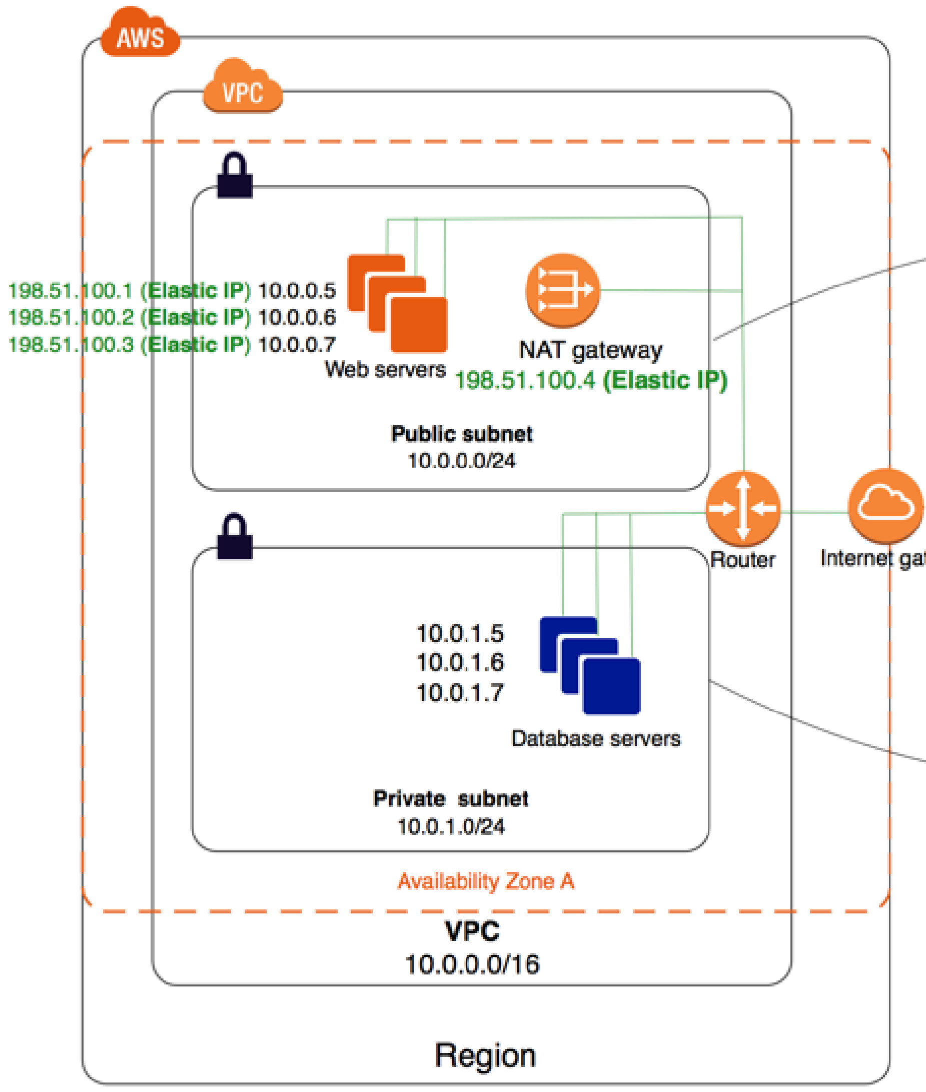


VPC Endpoint

- A VPC endpoint enables private connections between your VPC and supported AWS services (e.g. S3)
- They allow traffic to flow between a VPC and other services without ever leaving the Amazon network
- This introduces several benefits, not the least of which is improved network security



Tying this all together…



Custom route table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	*igw-id*

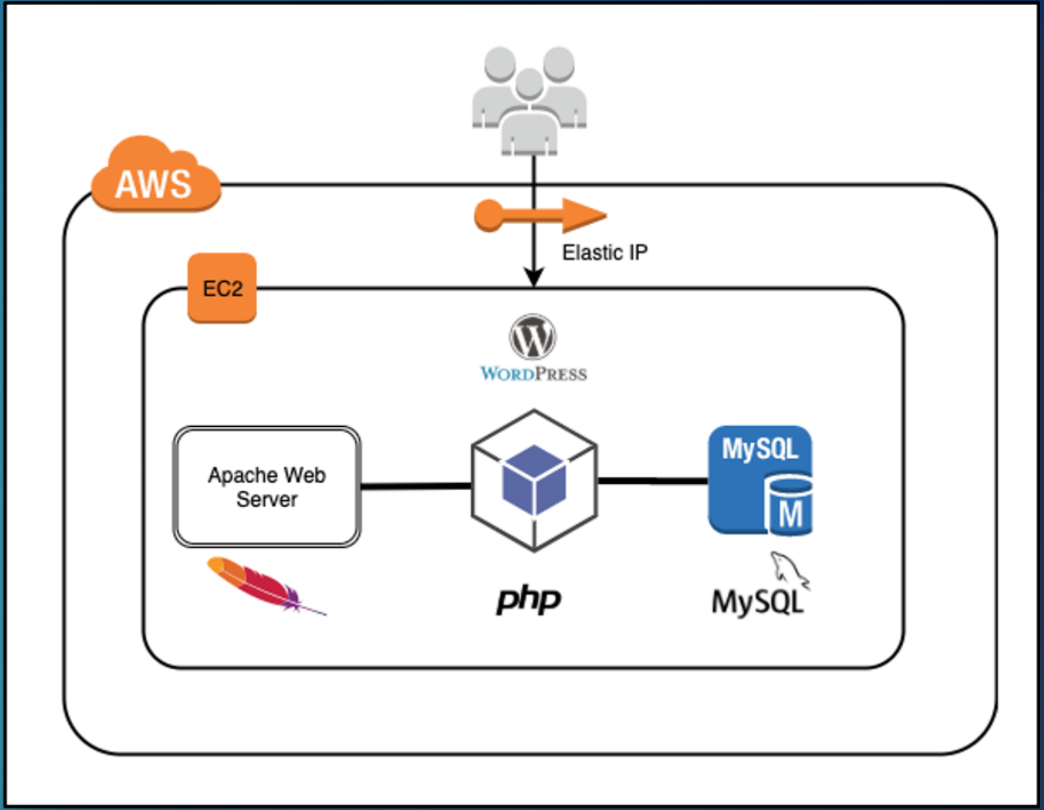
Main route table

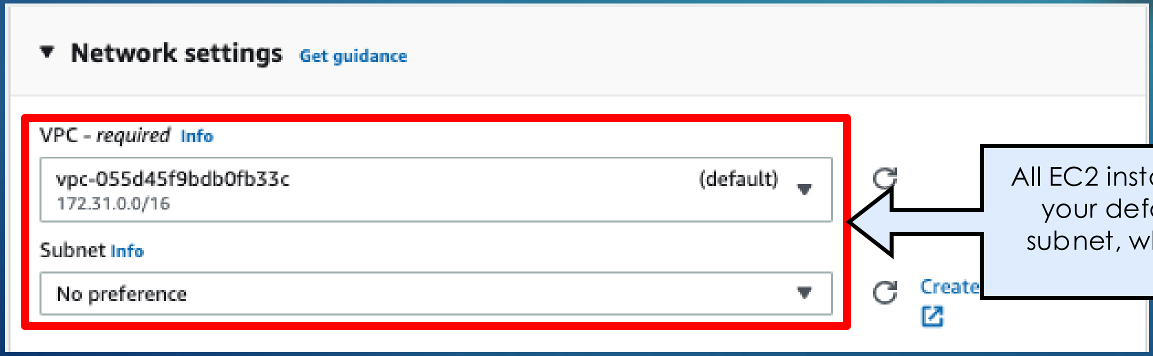
Destination	Target
10.0.0.0/16	local
0.0.0.0/0	*nat-gateway-id*

Source: https://gruntwork.io/guides/networking/how-to-deploy-production-grade-vpc-aws/#vpc_peering

VPC and Subnets

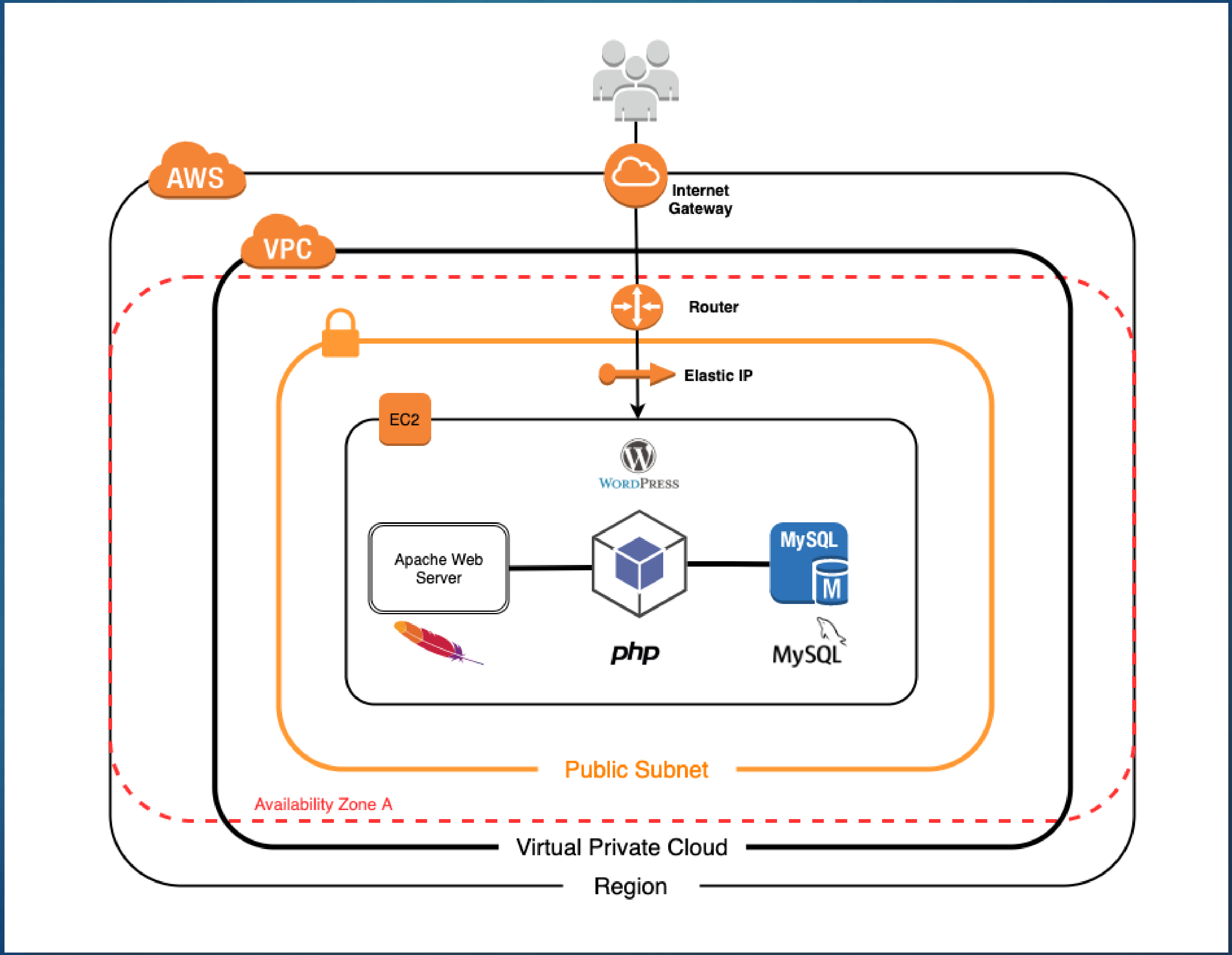
- For the WordPress EC2 instance you created, was it deployed in a VPC?
- For the WordPress EC2 instance you created, what subnet was it deployed to:
  - Public
  - Private
  - We haven't created it yet





All EC2 instances are launched in your default VPC and public subnet, which comes with your account

VPC and Subnets



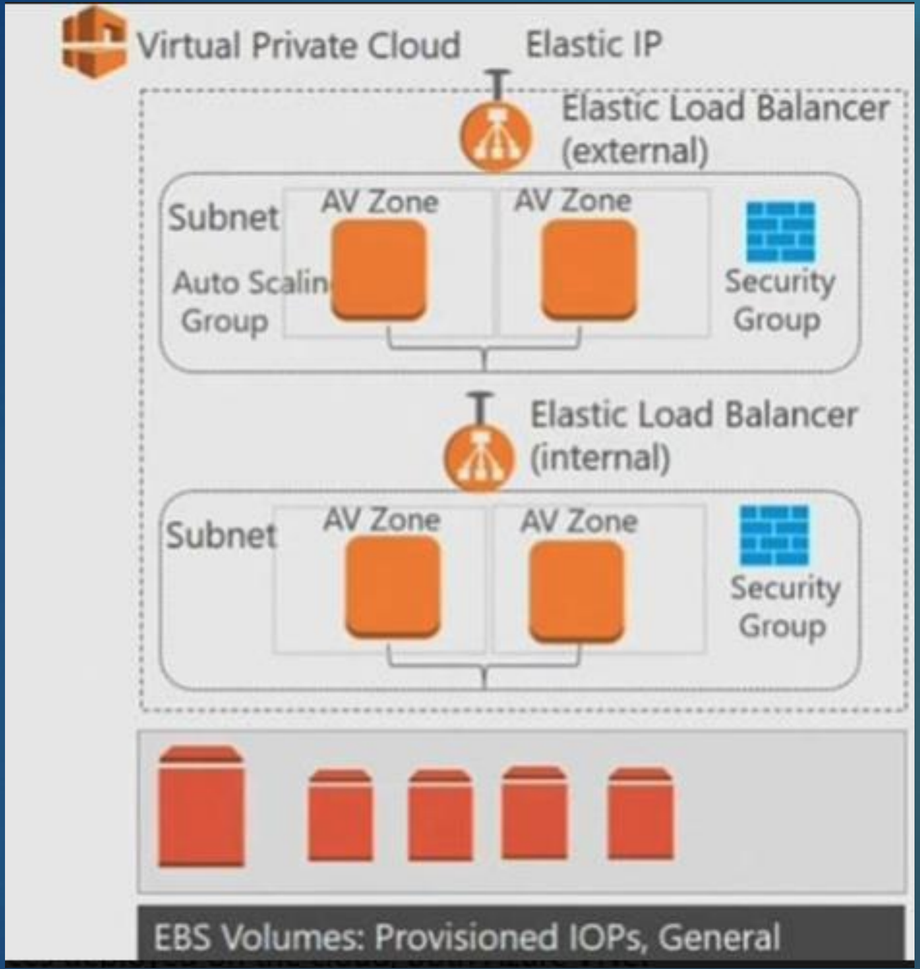
A VPC is located in a Region

Within a Region there are 2 or more Availability Zones

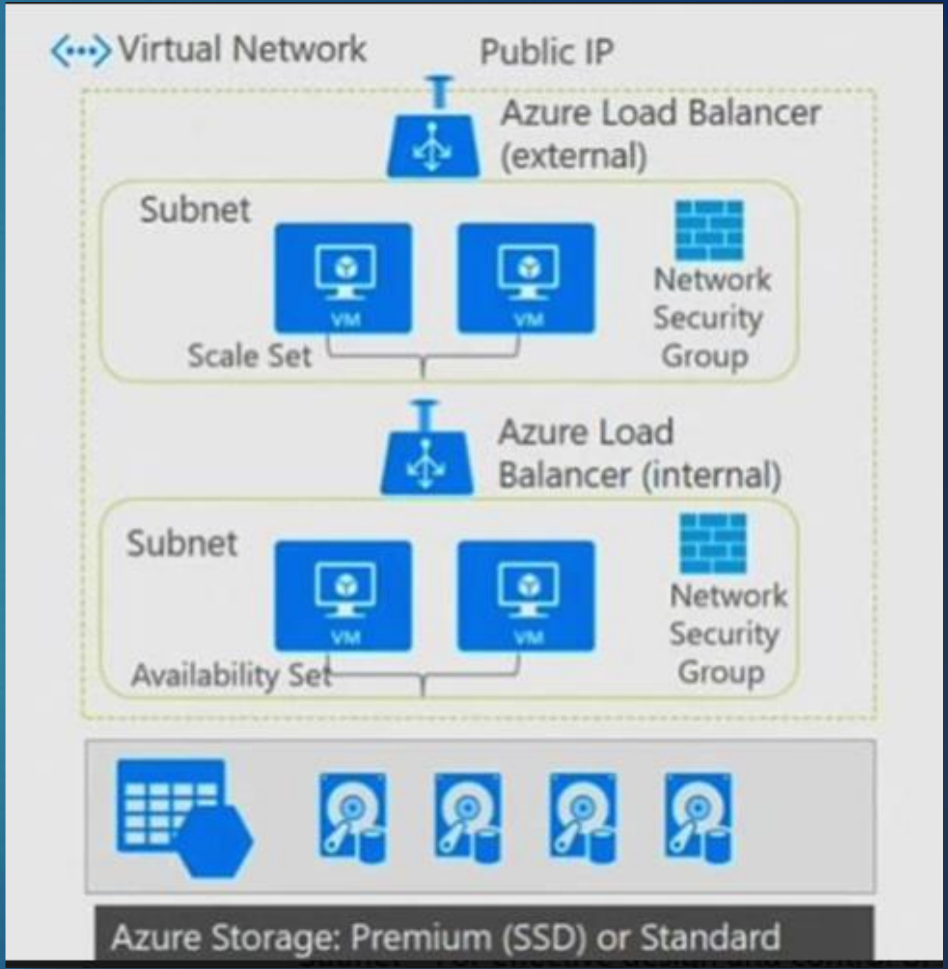
A Subnet is assigned to an Availability Zone

EC2

VPC and Subnets – AWS vs. Azure



AWS



Azure

Source: [Differentiating between Azure Virtual Network (VNet) and AWS Virtual Private Cloud (VPC)]

Class Activity

- Go to myCourses to Assignments > Activity – Creating a VPC and Subnets
  - Remember to take a screenshot when you are done